

University of Botswana — CSI132: Discrete Structures II

Tutorial 1: Proof Methods | Full Marking Key with Working

Proof Method Reference — Techniques Used in This Tutorial		
Method	Structure	When to Use
Direct Proof	Assume p . Use definitions/algebra to conclude q .	Straightforward implications; clear algebraic path.
Proof by Contraposition	Prove $\neg q \Rightarrow \neg p$ instead of $p \Rightarrow q$.	When the negation of the conclusion is easier to work with.
Proof by Contradiction	Assume $p \wedge \neg q$. Derive a contradiction $(r \wedge \neg r)$.	Negation of conclusion leads to an impossible statement.
Biconditional (Iff)	Prove $p \Rightarrow q$ AND $q \Rightarrow p$ separately.	Statements of the form "if and only if".
Key Definitions	n is even $\Leftrightarrow n = 2k$ for some integer k . n is odd $\Leftrightarrow n = 2k + 1$ for some integer k .	

Question 2 | Prove: The sum of two even integers is even.

PROOF METHOD: DIRECT PROOF

We use the definition of even integers directly. If m and n are even, write them in the form $2k$ and show the sum fits the same form.

DIRECT PROOF

ASSUME

Let m and n be even integers. By definition of even:

$$m = 2a \quad \text{and} \quad n = 2b \quad \text{for some integers } a, b \in \mathbb{Z}$$

COMPUTE THE SUM

$$m + n = 2a + 2b = 2(a + b)$$

CONCLUDE

Since $a + b$ is an integer (the integers are closed under addition), $m + n = 2(a + b)$ is even by definition.

CONCLUSION

The sum of any two even integers is even. ■

Question 3 | Prove: The additive inverse (negative) of an even number is even.

PROOF METHOD: DIRECT PROOF

The additive inverse of n is $-n$. We must show: if n is even, then $-n$ is even.

DIRECT PROOF

ASSUME

Let n be an even integer. By definition:

$$n = 2k \quad \text{for some integer } k \in \mathbb{Z}$$

COMPUTE THE ADDITIVE INVERSE

$$-n = -(2k) = 2(-k)$$

CONCLUDE

Since $-k$ is an integer (the integers are closed under negation), $-n = 2(-k)$ has the form $2 \times \text{integer}$, so $-n$ is even by definition.

CONCLUSION

The additive inverse of any even number is even. ■

Question 4 | Prove: The product of two odd numbers is odd.

PROOF METHOD: DIRECT PROOF

Write both odd numbers in the form $2k + 1$, multiply out, and show the product fits the same form.

DIRECT PROOF

ASSUME

Let m and n be odd integers. By definition:

$$m = 2a + 1 \quad \text{and} \quad n = 2b + 1 \quad \text{for some integers } a, b \in \mathbb{Z}$$

COMPUTE THE PRODUCT

$$\begin{aligned} m \cdot n &= (2a + 1)(2b + 1) \\ &= 4ab + 2a + 2b + 1 \\ &= 2(2ab + a + b) + 1 \end{aligned}$$

CONCLUDE

Let $c = 2ab + a + b$. Since $a, b \in \mathbb{Z}$, c is an integer, and:

$$m \cdot n = 2c + 1$$

This is of the form $2 \times \text{integer} + 1$, so $m \cdot n$ is odd by definition.

CONCLUSION

The product of any two odd integers is odd. ■

Question 5 | Prove: If m and n are integers and $m \cdot n$ is even, then m is even or n is even.

PROOF METHOD: PROOF BY CONTRAPOSITION

The statement is: $m \cdot n$ even $\Rightarrow (m \text{ even} \vee n \text{ even})$.

The contrapositive is: $\neg(m \text{ even} \vee n \text{ even}) \Rightarrow m \cdot n$ is not even, i.e. $(m \text{ odd} \wedge n \text{ odd}) \Rightarrow m \cdot n$ odd. This is exactly what we proved in Question 4!

PROOF BY CONTRAPOSITION

STATE THE CONTRAPOSITIVE

We prove the logically equivalent contrapositive:

If m is odd AND n is odd, then $m \cdot n$ is odd.

ASSUME

Suppose m and n are both odd. By definition:

$$m = 2a + 1, \quad n = 2b + 1 \quad \text{for some } a, b \in \mathbb{Z}$$

COMPUTE

$$m \cdot n = (2a + 1)(2b + 1) = 4ab + 2a + 2b + 1 = 2(2ab + a + b) + 1$$

CONCLUDE THE CONTRAPOSITIVE

$m \cdot n = 2(2ab + a + b) + 1$ is odd. So the contrapositive holds.

Since $p \Rightarrow q \equiv \neg q \Rightarrow \neg p$, the original statement is proved.

CONCLUSION

If $m \cdot n$ is even, then m is even or n is even. ■

Question 6 | Prove: If $3n + 2$ is even, then n is even.

Part (a) — Proof by Contraposition

CONTRAPOSITIVE

Original: $(3n + 2 \text{ even}) \Rightarrow (n \text{ even})$. Contrapositive: $(n \text{ odd}) \Rightarrow (3n + 2 \text{ odd})$.

PROOF BY CONTRAPOSITION

ASSUME

Suppose n is odd. Then $n = 2k + 1$ for some integer k .

COMPUTE $3n + 2$

$$\begin{aligned} 3n + 2 &= 3(2k + 1) + 2 \\ &= 6k + 3 + 2 \\ &= 6k + 5 \\ &= 2(3k + 2) + 1 \end{aligned}$$

CONCLUDE

Since $3k + 2$ is an integer, $3n + 2 = 2(3k + 2) + 1$ is odd. The contrapositive holds.

Therefore, by contraposition, if $3n + 2$ is even then n is even.

CONCLUSION (A)

By contraposition: if $3n + 2$ is even, then n is even. ■

Part (b) — Proof by Contradiction

CONTRADICTION SETUP

Assume both the hypothesis AND the negation of the conclusion are true simultaneously, then derive a contradiction.
Assume: $3n + 2$ is even AND n is odd.

PROOF BY CONTRADICTION

ASSUME FOR CONTRADICTION

Suppose $3n + 2$ is even and n is odd (i.e., n is NOT even).
Since n is odd: $n = 2k + 1$ for some integer k .

COMPUTE $3n + 2$ UNDER THIS ASSUMPTION

$$3n + 2 = 3(2k + 1) + 2 = 6k + 5 = 2(3k + 2) + 1$$

So $3n + 2$ is **odd**.

IDENTIFY THE CONTRADICTION

We assumed $3n + 2$ is even, but we derived $3n + 2$ is odd. A number cannot be both even and odd simultaneously. $\Rightarrow$ Contradiction! \lightning

Our assumption that n is odd must be false. Therefore n is even.

CONCLUSION (B)

By contradiction: if $3n + 2$ is even, then n is even. ■

Question 7 | Prove: n is even if and only if $7n + 4$ is even, for n a positive integer.

PROOF METHOD: BICONDITIONAL — PROVE BOTH DIRECTIONS

We must prove:

$(\Rightarrow)$: If n is even, then $7n + 4$ is even.

$(\Leftarrow)$: If $7n + 4$ is even, then n is even.

Direction $(\Rightarrow)$: n even $\Rightarrow 7n + 4$ even — Direct Proof

DIRECT PROOF

ASSUME

Let n be even. Then $n = 2k$ for some integer k .

COMPUTE

$$7n + 4 = 7(2k) + 4 = 14k + 4 = 2(7k + 2)$$

CONCLUDE

Since $7k + 2$ is an integer, $7n + 4 = 2(7k + 2)$ is even. ✓

Direction ($\Leftarrow$): $7n + 4$ even $\Rightarrow n$ even — Proof by Contraposition

PROOF BY CONTRAPOSITION

STATE THE CONTRAPOSITIVE

We prove: if n is odd, then $7n + 4$ is odd.

ASSUME

Let n be odd. Then $n = 2k + 1$ for some integer k .

COMPUTE

$$7n + 4 = 7(2k + 1) + 4 = 14k + 7 + 4 = 14k + 11 = 2(7k + 5) + 1$$

CONCLUDE

Since $7k + 5$ is an integer, $7n + 4 = 2(7k + 5) + 1$ is odd. The contrapositive holds, so the backward direction is proved. $\checkmark$

CONCLUSION

Both directions have been proved. Therefore n is even $\Leftrightarrow 7n + 4$ is even. $\blacksquare$

Question 8 | Prove: $m^2 = n^2$ if and only if $m = n$ or $m = -n$.

PROOF METHOD: BICONDITIONAL — ALGEBRAIC + CASE ANALYSIS

The key tool is the factorisation:

$$m^2 - n^2 = (m - n)(m + n)$$

A product of two integers equals zero if and only if at least one factor is zero. We prove both directions: ($\Rightarrow$) and ($\Leftarrow$).
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Direction ($\Rightarrow$): $m^2 = n^2 \Rightarrow m = n$ or $m = -n$ — Direct Proof

DIRECT PROOF WITH CASE ANALYSIS

ASSUME

Suppose $m^2 = n^2$.

REARRANGE

$$m^2 - n^2 = 0 \implies (m - n)(m + n) = 0$$

APPLY ZERO-PRODUCT PROPERTY

Since $(m - n)(m + n) = 0$, at least one factor must be zero:

$$m - n = 0 \quad \text{or} \quad m + n = 0$$

INTERPRET EACH CASE

Case 1: $m - n = 0 \Rightarrow m = n$. $\checkmark$

Case 2: $m + n = 0 \Rightarrow m = -n$. $\checkmark$

Therefore $m^2 = n^2 \Rightarrow (m = n \vee m = -n)$. $\checkmark$

Direction ($\Leftarrow$): $m = n$ or $m = -n \Rightarrow m^2 = n^2$ – Direct Proof

DIRECT PROOF – TWO CASES

CASE 1: SUPPOSE $M = N$

$$m^2 = m \cdot m = n \cdot n = n^2 \checkmark$$

CASE 2: SUPPOSE $M = -N$

$$m^2 = (-n)^2 = (-n)(-n) = n^2 \checkmark$$

In both cases, $m^2 = n^2$. $\checkmark$

CONCLUSION

Both directions have been proved. Therefore $m^2 = n^2 \Leftrightarrow (m = n \text{ or } m = -n)$. $\blacksquare$